FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Advanced)-2020

PAPER – 1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. A

Sol. $\vec{V}_{sn} = (v_0 + \alpha x)\hat{i} + vj$

time taken, $t = \frac{L}{V}$

Now, $\frac{dx}{dt} = (v_0 + \alpha x) \Rightarrow \int_0^x \frac{dx}{v_0 + \alpha x} = \int_0^t dt$

$x = \frac{v_0}{\alpha}\left[e^{\frac{L\alpha}{v}} - 1\right]$

2. B

Sol. $V = \frac{\int Fdt}{m} = \frac{\int iLBdt}{m}$

3. A

Sol. $4C_0T_0^x = C_0(2T_0)^x$

$\therefore x = 2$

Now, $CT^2 = 4C_0T_0^2$

$C = \frac{4C_0T_0^2}{T^2}$

$\int dq = \int_{T_0}^{2T_0} nCdT$

$q = 2nC_0T_0$

4. B

Sol. $V = \omega R - v\cos\theta$

$$V = \frac{\omega R}{1+\cos\theta}$$

5. B, C, D

Sol. Use concept of motion

6. A, B, C

Sol. From A to B,

$$mg(5r - r)\sin 30^0 = \frac{1}{2}mv_0^2 + \frac{1}{2}I\omega^2$$

$$\omega = \sqrt{\frac{20g}{7r}}$$

Apply energy equation between B and E

$$h = \frac{31r}{7}$$

7. A, C

Sol. For process AC;

$P = a - bv \ \ [a, b = \text{constant}]$

$\therefore (a - bv)(v) = RT$

$\therefore T = \frac{1}{R}(av - bv^2)$

$\therefore \frac{dT}{dv} = \frac{a}{R} - \frac{2bv}{R}$

$0 = \frac{a}{R} - \frac{2bv}{R} \Rightarrow v = \frac{a}{2b}$

Also,

$$\left(\frac{d^2T}{dv^2}\right)_{\frac{a}{2b}} < 0$$

$\therefore$ For AC, T first increases and then decreases.

For AD, T $\downarrow$, $\therefore$ K.E $\downarrow$

8. A, B

Sol. Concept based

9. A, D

Sol. $q_1 + q_2 = 2Q_0 \qquad .....(i)$

$C_1 = \frac{A\varepsilon_0}{d_0 + vt}, C_2 = \frac{A\varepsilon_0}{d_0 vt}$

$\therefore \frac{C_1}{C_2} = \frac{d_0 - vt}{d_0 + vt} = \frac{q_1}{q_2} \qquad .....(ii)$

$\therefore$ from (i) and (ii),

$$q_2 = \frac{q_0(d_0 + vt)}{d_0}$$

and

$$I = \frac{dq_2}{dt} = \frac{q_0 v}{d_0}$$

10. A, B

Sol. Use momentum conservation and impulse momentum equation.

11. B, C, D

Sol. For $n_1 \to n_2$

$$\frac{1}{\lambda_1} = A\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) \quad .....(i)$$

and For $n_2 \to (n_2 + 1)$

$$\frac{1}{\lambda_2} = A\left(\frac{1}{n^2} - \frac{1}{(n_2+1)^2}\right) \quad .....(ii)$$

and it is given that,

$\lambda_1 = 102.8$ nm

$\lambda_2 = 97.5$ nm

solving (i) and (ii)

$n_1 = 1$, then $n_2 = 3$ and $n_{2+1} = 4$

and $A = 2.174 \times 10^{-18} J = 13.6$ ev

12. A, B, C

Sol. Path difference (Δx) between two consecutive wave = $d\sin\theta = d\left(\frac{Y}{L}\right)$

For minimum, phase difference between two consecutive wave is $\frac{2\pi}{3}$ or $\frac{4\pi}{3}$.

Hence,

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta x$$

$$\therefore y = \frac{\lambda L}{3d}, \frac{2\lambda L}{3d}, \frac{4\lambda L}{3d} .....$$

SECTION – C

13. 00001.00

Sol. Use the concept of motional emf.

14. 00004.00

Sol. Time taken to reach the middle of the river form the river bank, $t = \frac{50}{5} = 10$ second.

Now, $v_x = \left(\frac{2}{10}\right)t$

$$\int_0^x dx = \frac{1}{5}\int_0^{10} t\,dt$$

$x = 10$ m

Hence total difference = $10 \times 2 = 20$ m.

15. 00003.00

Sol. $1.\sin 45^0 = \sqrt{2}\ \sin r$

$r = 30^0$

Now,

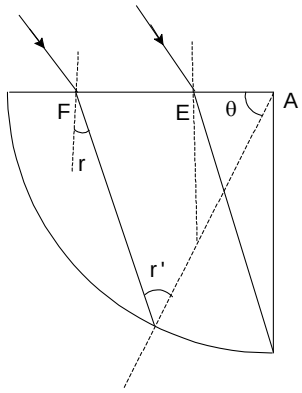
At G,

$r' = \left(120^0 - \theta\right)$

$\therefore$ For TIR,

$\sqrt{2}\sin\left(120^0 - \theta\right) = 1.\sin 90^0$

$\theta = 75^0$



Now

$AE = \dfrac{R}{\sqrt{3}},\ AF = \dfrac{\sqrt{2}R}{\sqrt{3}}$

$\therefore EF = \dfrac{\left(\sqrt{2}-1\right)R}{\sqrt{3}}$

Hence,

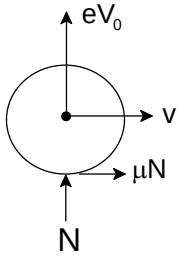
$\text{Area, } a = \dfrac{\left(\sqrt{2}-1\right)}{\sqrt{3}} \times \dfrac{\left(\sqrt{2}+1\right)R}{\sqrt{3}}$

$= \dfrac{1}{2}R^2 = \dfrac{9}{3} = 3$

16. 00007.00

Sol. $\int N dt = mev_0 - \left(-mv_0\right) \quad ....(i)$

$\int \mu N dt = mv \quad .....(ii)$



from (i) and (ii)

$v = v_0\left(1+e\right)\mu$

Now,

$t = \dfrac{2ev_0}{g}$

$\text{distance, } d = vt = 7$

17. 00007.00

Sol. $P_{eq} = 2P_L + P_m = \dfrac{2}{2R} + \dfrac{2}{R}$

$P_{eq} = \dfrac{3}{R}$

Now,

$-\dfrac{2}{R_{meq}} = \dfrac{3}{R}$

$\left|R_{meq}\right| = \dfrac{2R}{3}$

Hence, Image formed by the plane mirror is at a distance of $\frac{2R}{3} = 28$ cm.

$\therefore d + 21 = 28$

$d = 7$ cm

18. 00002.00

Sol. At time t,

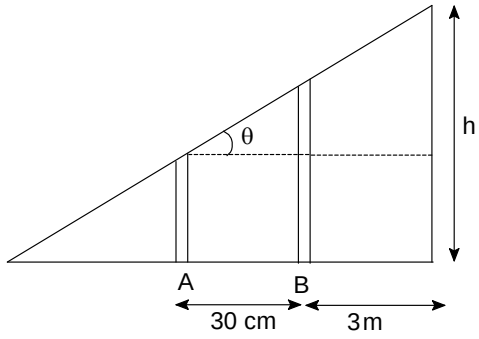
$h_A = 30 - \frac{30}{4}t,\ h_B = 30 - \frac{30}{6}t$

Now,

$\tan\theta = \frac{h_B - h_A}{30} = \frac{h}{330}$

$\therefore h = 30 - 20t$

$\therefore \frac{dh}{dt} = 20$ cm / hr $= 2$ decimeter / hr



Chemistry

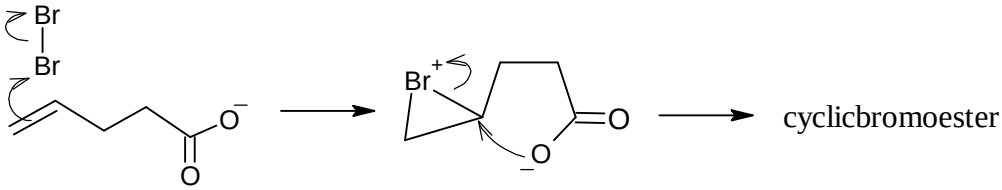
PART – II

SECTION – A

19. D

Sol. White precipitate = AgCl
Yellow precipitate = AgI
Black precipitate = Ag_2O

20. C

Sol. 

Intermolecular attack of bromide ion on the bromonium ion does not compete with the intramolecular cyclisation step.

21. D

Sol. The brown ppt. is $Fe(OH)_3$ on drying and heating it becomes Fe_2O_3 carbon monoxide will reduce it to iron metal.

$$Fe_2O_3 + 3CO \xrightarrow{700^0 c} 2Fe + 3CO_2$$

22. A

Sol. $\Delta[I^-] = [I^-]_{final} - [I^-]_{initial}$

$= 0.23 - 0.25 = -0.02 mol L^{-1}$.

$\Delta t = 10 - 0 = 10 \min$

$$-\frac{\Delta[I^-]}{\Delta t} = \frac{-(-0.02)}{10} = 0.002 mol L^{-1} \min^{-1}$$

$\therefore$ Rate of appearance of $I_2 = -\frac{1}{2}\frac{\Delta[I^-]}{\Delta t} = \frac{0.002}{2} = 1\times10^{-3} mol\, L^{-1} \min^{-1}$

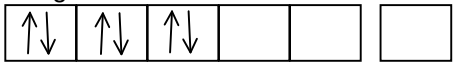
23. BC

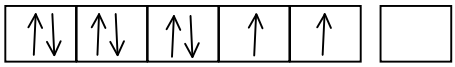
Sol. $3-\text{methyl}-1-\text{butyne} \xrightarrow[H_2SO_4]{Hg^{+2}} 3-\text{methyl}-2-\text{butanone} \xrightarrow{NaBH_4} 3-\text{methyl}-2-\text{butanol}$

24. ABD

Sol. (A) The magnitude of splitting energy increases down the group. Thus
Pt^{2+} – $5d^8$, Pd^{2+} – $4d^8$ and Ni^{2+} – $3d^8$.

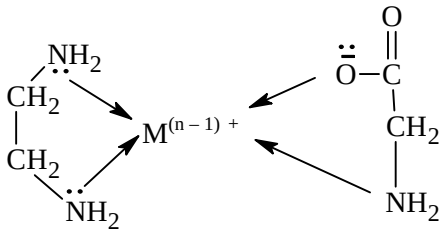
(B) NH_3 is a borderline ligand which forces pairing of electrons in Co^{3+}, whereas H_2O is a weak-field ligand and it can not force the pairing in Ni^{2+}.

Co^{3+} in $[Co(NH_3)_6]^{3+}$ – $3d^6, 4s^0$ 

Ni^{2+} in $[Ni(H_2O)_6]^{2+}$ – $3d^8, 4s^0$ 

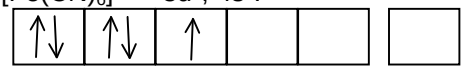
So, there is no unpaired electron in $[Co(NH_3)_6]^{3+}$ whereas there are unpaired electron in $[Ni(H_2O)_6]^{2+}$. Thus $[Co(NH_3)_6]^{3+}$ is colourless and $[Ni(H_2O)_6]^{2+}$ is coloured due to d–d transition.

(C)



As en is symmetrical ligand thus there is no geometrical isomerism.

(D) In $K_3[Fe(CN)_6]$, the oxidation state of Fe is +3.
$[Fe(CN)_6]^{3-}$ – $3d^5$, $4s^0$.



Number of unpaired electron = 1

$\therefore \mu = \sqrt{n(n+2)}$ B.M. = $\sqrt{1(1+2)}$ B.M. = $\sqrt{3}$ B.M.

25. ABD

Sol. $r_{K^+} > r_{Na^+}$
$\therefore$ K_2CO_3 is more ionic.
$(CN)_2 + NaOH \rightarrow NaCN + NaOCN + H_2O$
$Br_2 + OH^- \rightarrow Br^- + BrO^- + H_2O$.

26. AB

Sol. Na_2O has antifluorite structure, in which O^{-2} ions form ccp array and Na^+ ions occupy all tetrahedral voids.

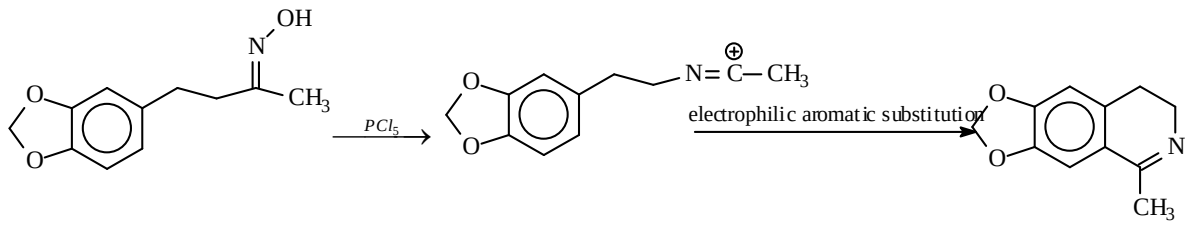
27. ABCD

Sol. Alkyl isocyanate is $R-N=C=O$

(A) Curtius rearrangement (B) Schmidt rearrangement
(C) Lossen rearrangement (D) Hofmann rearrangement

28. ABC

Sol. a and b are obtained by normal Beckmann rearrangement
mechanism for product 'c' is



29. ABC

Sol. $CH_3COONa \longrightarrow C_2H_6$ (major)

CH_2COOK
$|$ $\longrightarrow CH_2 = CH_2$
CH_2COOK

$$\begin{matrix} HC-COOK \\ \| \\ HC-COOK \end{matrix} \longrightarrow \quad CH \equiv CH$$

30. BCD

Sol. For $C(\text{graphite}) \xrightarrow[\text{high P}]{\text{high T}} C(\text{diamond})$; $\Delta H = +ve$ and $\Delta S = -ve$

SECTION – C

31. 00003.00

Sol. $ROH \xrightarrow{CH_3COCl} ROCOCH_3$

Here H is replaced by $COCH_3$ i.e. one H of mass 1 is replaced by one $COCH_3$ of mass 43.

$\therefore$ Mass increased is 42.

$\therefore$ Mass of acylated product is = $92 \times \frac{237}{100} = 218$

$\therefore$ increase in mass = 218 – 92 = 126

$\therefore$ number of OH = 126/42 = 3

32. 00005.00

Sol. $2I^- \rightarrow I_2 + 2e^-$

$I_2 + 2S_2O_3^- \rightarrow 2I^- + S_4O_6^{-2}$

$2I^- + MnO_4^- \rightarrow I_2 + Mn^{+2}$

No. of eq. of KI electrolysed

= no. of eq. of I_2 produced

= no. of eq. of $Na_2S_2O_3 = 0.25 \times 37.2 \times 1 \times 10^{-3}$

$= 9.3 \times 10^{-3}$

No. of eq.= $\frac{It}{96500} = \frac{I \times 3 \times 60}{96500} = 9.3 \times 10^{-3}$

$\Rightarrow$ I = 5 A

33. 00363.52

Sol. According to Raoult's law:

$P_T = p_1^0 x_1 + p_2^0 x_2$

$p_1 + p_2 = p_1^0(0.16) + p_2^0(0.84) = 760mm$

Also $p_1^0 - p_2^0 = 472mm$

$\therefore \quad 760 = p_1^0(0.16) + (p_1^0 - 472)(0.84)$

$760 = 0.16p_1^0 + 0.84p_1^0 - 396.48$

$\therefore \quad p_1^0 = 363.52$ mm

34. 00000.00

Sol. no movement of D.P. towards any electrode in during electro-osmosis

35. 00006.40

Sol. $CaC_2O_4 \rightleftharpoons \underset{S}{Ca^{++}} + \underset{S}{C_2O_4^{--}}$

$K_{sp} = S \times S = S^2$

$s = \sqrt{K_{sp}} = (2.5 \times 10^{-9})^{1/2} = 5 \times 10^{-5}$ mol litre^{-1} $\therefore \frac{w}{128} = 5 \times 10^{-5}$

$\therefore$ w = 6.4×10^{-3}g = 6.4 mg

36. 00069.16

Sol. Let the mass of acetaldehyde is *x* g, then mass of ethanol is (0.535 – *x*)g. When acetaldehyde is heated with Fehling solution, it gives 1.2 g of a red precipitate.

$CH_3CHO + 2CuO \rightarrow CH_3COO^- + \underset{\text{Red ppt.}}{Cu_2O}$

$\because$143 g of Cu_2O is coming from = 44 g of CH_3CHO

$\therefore$1.2 g of Cu_2O is coming from $\frac{44}{143} \times 1.2$ = 0.37 g of CH_3CHO

% of CH_3CHO = $\frac{0.37}{0.535} \times 100$ = 69.16 %

Mathematics

PART – III

SECTION – A

37. A

Sol. $\dfrac{1\times(2a)+2\left(\dfrac{b}{2}\right)+2\left(\dfrac{3c}{2}\right)}{5}\geq\left((2a)\left(\dfrac{b}{2}\right)^2\left(\dfrac{3c}{2}\right)^2\right)^{1/5}$

$\dfrac{8}{9\times5^{-5}}\geq ab^2c^2$

38. D

Sol. $P(T)=\underbrace{\dfrac{1}{3}\cdot\dfrac{1}{4}}_{a}+\underbrace{\dfrac{1}{3}\cdot\dfrac{1}{2}}_{b}+\underbrace{\dfrac{1}{3}\cdot\dfrac{4}{5}}_{c}$

So, required probability= $\dfrac{1}{a+b+c}\left(\dfrac{a\cdot1}{4}+\dfrac{b\cdot1}{2}+\dfrac{c\cdot4}{5}\right)=\dfrac{381}{620}$

39. A

Sol. $I=\int\dfrac{\left(\tan x+1+\tan^2 x\right)-\left(1+\tan^2 x\right)}{\tan x+1+\tan^2 x}dx=x-\int\dfrac{1+\tan^2 x}{1+\tan x+\tan^2 x}dx$

$z=\tan x \qquad dz=\sec^2 x dx$

$I=x-\int\dfrac{dt}{\left(t+\dfrac{1}{2}\right)^2+\left(\dfrac{\sqrt{3}}{2}\right)^2}=x-\dfrac{2}{\sqrt{3}}\tan^{-1}\left(\dfrac{2\tan x+1}{\sqrt{3}}\right)+C$

40. A

Sol. $x=\dfrac{ab}{t}\Rightarrow dx=\dfrac{ab}{t^2}(-dt)$

$I=\int_b^a\dfrac{e^{\frac{b^2}{t^2}}-e^{\frac{t^2}{a^2}}}{\dfrac{ab}{t}}\left(-\dfrac{ab}{t^2}\right)dt=-I$

$\therefore 2I=0 \qquad I=0$

41. BC

Sol. $\left(x^2-1\right)\left(x^2-(a+2)x+3\right)=0$

Since $x^2-(a+2)x+3=0$ must have at least one positive real roots.

$\because \alpha\beta=3$ both root must be positive

$\alpha+\beta=a+2>0\Rightarrow a>-2\ \&\ D\geq0\Rightarrow(a+2)^2-12\geq0 \qquad a\geq2\left(\sqrt{3}-1\right)$

42. AB

Sol. $m=\dfrac{2m_1}{1-m_1^{\,2}}$

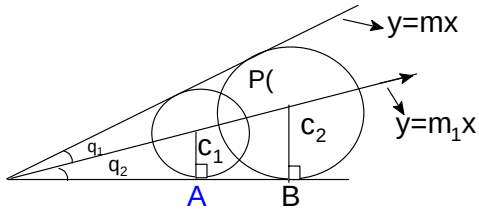
$O_1P^2=r_1^{\,2}=m_1^{\,2}\alpha^2$

$(9-\alpha)^2+(6-m_1\alpha)^2=m_1^{\,2}\alpha^2$

$\alpha^2-6(2m_1+3)\alpha+117=0$

$\alpha\beta=117 \qquad r_1r_2=m_1^{\,2}\alpha\beta=68$

$\therefore m=\pm\dfrac{12\sqrt{221}}{49}$



43. BD

Sol. $\dfrac{\left(\dfrac{\sqrt{3}-1}{2\sqrt{2}}\right)}{\sin x}+\dfrac{\left(\dfrac{\sqrt{3}+1}{2\sqrt{2}}\right)}{\cos x}=2\Rightarrow\dfrac{\sin\left(\dfrac{\pi}{12}\right)}{\sin x}+\dfrac{\cos\left(\dfrac{\pi}{12}\right)}{\cos x}=2$

44. AD

Sol. ∴ Given

$\left(\vec{a}+\vec{b}\right)\times\left(4\hat{i}+\lambda\,\hat{j}+2\hat{k}\right)=0$

$\left|\vec{a}+\vec{b}\right|\left|-3\hat{i}+2\,\hat{j}+\lambda\,\hat{k}\right|\cos\theta=10$

θ is the angle between $\left(4\hat{i}+\lambda\,\hat{j}+2\hat{k}\right)$ & $\left(-3\hat{i}+2\,\hat{j}+\lambda\,\hat{k}\right)$

$\therefore 7\lambda^2-48\lambda+52=0$

45. AC

Sol. $f'(x)=\lim\limits_{h\to0}\dfrac{f(x+h)-f(x)}{h}=\lim\limits_{h\to0}\dfrac{f\left(\dfrac{2x+2h}{2}\right)-f(x)}{h}$

$$= \lim_{h\to 0} \frac{f(2x)+f(2h)-2f(x)}{2h}$$

$$Let\ x = 2x\ \&\ y = 0 \qquad f(x) = \frac{f(2x)+f(0)}{2} = \frac{f(2x)+1}{2}$$

$$= \lim_{h\to 0} \frac{2f(x)-1+2f(2h)-2f(x)}{2h} = \lim_{h\to 0} \frac{f(2h)-1}{2h}$$

$$= \lim_{h\to 0} \frac{f(2h)-f(0)}{2h-0} = f'(0) = -1 \qquad \therefore f'(x) = -1$$

46. A

Sol. Use homogenization method

47. ABC

Sol. Use matrix property

48. ACD

Sol. $9x^3 - 7x + 6 = 0$

$y = 3x + 2$

$$x = \frac{y-2}{3}$$

$$9\left(\frac{y-2}{3}\right)^3 - 7\left(\frac{y-2}{3}\right) + 6 = 0$$

$$(y-2)^3 - 7(y-2) + 18 = 0$$

$$y^3 - 6y^2 + 5y + 24 = 0$$

SECTION – C

49. 00007.50

Sol. Direction ratio of PQ

$$(-3r_2 - 3r_1 - 6,\ 2r_2 + r_1 + 6,\ 4r_2 - r_1 - 2)$$

Now, $-3(-3r_2 - 3r_1 - 6) + 2(2r_2 + r_1 - 16) + 4(4r_2 - r_1 - 2) = 0$

$\therefore 7r_1 + 29r_2 = 22 \qquad ---(1)$

$3(-3r_2 - 3r_1 - 6) - (2r_2 + r_1 - 16) + (4r_2 - r_1 - 2) = 0$

$7r_2 + 11r_1 = -4 \qquad ---(2) \qquad \therefore r_1 = -1,\ r_2 = 1$

$a = -3,\ b = 2,\ c = 5$

$$-a + b + \frac{c}{2} = 3 + 2 + 25 = 7.5$$

50. 00002.50

Sol. $12\sin x + 5\cos x = 2(y-2)^2 + 13$

$by\ LHS \le 13, \quad RHS \ge 13$

$\therefore y = 2\ \ \therefore 12\sin x + 5\cos x = 13\cos(x-\alpha) = 1\ x = 2x\pi + \alpha$

51. 00001.98

Sol. $p(x)=(x-1)(x-2)(x-3)(x-\alpha)+10x$

$\therefore\ p(12)+p(-8)=19800+40$

$$\frac{p(12)+p(-8)-10}{10000}=1.98$$

52. 00102.53

Sol. Let $\dfrac{\tan 8^\circ}{1-3\tan^2 8}=\dfrac{1}{8}\left[\dfrac{8\tan 8^\circ}{1-3\tan^2 8}+\tan 8^\circ-\tan 8^\circ\right]=-\dfrac{1}{8}\tan 8^\circ+\dfrac{3}{8}\tan 24^\circ$

$\therefore$ Solving LHS

$$\frac{81}{8}\tan 108^\circ-\frac{1}{8}\tan 8^\circ=x^2+y^2$$

$$x^2+y^2=\frac{(81)^2+(1)^2}{64}=102.53$$

53. 00000.50

Sol. $$\prod_{K=0}^{n}{}^nC_K=\prod_{K=1}^{n}\frac{n!}{K!(n-K)!}=\frac{(n!)^{n+1}}{(1!2!...3!)^2}$$

$$=\left(\prod_{K=1}^{n}(n+1-K)^{n+1-2K}\right)$$

$$=\prod_{K=1}^{n}\left(\frac{n+1-K}{n+1}\right)^{n+1-2K}\quad \sin ce\ \sum(n+1-2K)=0$$

Taking log and limit we get

$$\frac{1}{n}\sum_{K=1}^{n}\left(1-\frac{2K}{n+1}\right)\ln\left(1-\frac{K}{n+1}\right)$$

$$=\int_0^1(1-2x)\ln(1-x)\,dx=\frac{1}{2}$$

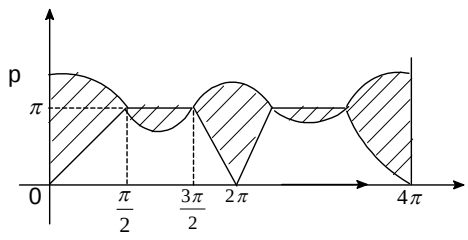
54. 00017.86

Sol. $f(x)=2x;\ 0\le x\le\dfrac{\pi}{2}$

$=\pi;\ \dfrac{\pi}{2}<x\le\dfrac{3\pi}{2}$

$=4\pi-2x;\ \dfrac{3\pi}{2}<x\le 2\pi$

Required Area



$$= 4\int_{0}^{\pi/2}\left(\pi + \cos x - 2x\right)dx + 2\int_{\pi/2}^{3\pi/2}\left(\pi - \left(\pi + \cos x\right)\right)dx$$

$$= \pi^2 + 8 = \left(3.14\right)^2 + 8 = 17.86$$